Pagination Unit Testing

Case 1 : In DoctorList

DoctorService :

```
1 usage  new *
public Page<Doctor> getAllDoctors(Integer pageNo, Integer pageSize, String SortBy) {
    Pageable pageable = PageRequest.of(pageNo, pageSize, Sort.by( ...properties: "id"));
    Page<Doctor> pagedResult = doctorRepository.findAll(pageable);
    return pagedResult;
}
```

DoctorRepository :

```
new *
Page<Doctor> findAll(Pageable pageable);
```

DoctorController :

```
no usages   reddy88 +1 *
@GetMapping("/doctor")
public ResponseEntity<Page<Doctor>> getAllDoctors(
        @RequestParam(defaultValue = "0") Integer pageNo,
        @RequestParam(defaultValue = "5") Integer pageSize,
        @RequestParam(defaultValue = "id") String sortBy) {
    Page<Doctor> doctors = doctorService.getAllDoctors(pageNo, pageSize, sortBy);
    return ResponseEntity.ok()
            .header( headerName: "X-Total-Count", String.valueOf(doctors.getTotalElements()))
            .header( headerName: "X-Page-No", String.valueOf(doctors.getNumber()))
            .header( headerName: "X-Page-Size", String.valueOf(doctors.getSize()))
            .body(doctors);
}
```

Response :

(http://localhost:8090/api/v1/doctors?pageNo=0&pageSize=2&sortBy=id)


```
{
    "content": [
        {
            "createdDate": null,
            "createdBy": "nasreen",
            "lastModifiedDate": null,
            "lastModifiedBy": "nasreen
",
            "id": 1,
            "doctorSpecialization": {
                "createdDate": null,
                "createdBy": null,
```

```
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "id": 1,
                "specializationName": "Dentist",
                "specializationDescription": "asdsdsfg"
            },
            "hospital": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "hospitalId": 1,
                "name": "Swasti",
                "address": "patna"
            },
            "fullName": "Ayushi",
            "registrationNumber": "1",
            "mobile": 1234567890,
            "email": "ab@gmail.com",
            "gender": "female",
            "rating": 5
        },
        {
            "createdDate": "2023-04-19T10:30:13.000+00:00",
            "createdBy": null,
            "lastModifiedDate": "2023-04-19T10:30:13.000+00:00",
            "lastModifiedBy": null,
            "id": 2,
            "doctorSpecialization": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "id": 1,
                "specializationName": "Dentist",
                "specializationDescription": "asdsdsfg"
            },
            "hospital": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "hospitalId": 2,
                "name": "ram",
                "address": "Delhi"
            },
            "fullName": "Ayushi",
            "registrationNumber": "A1",
            "mobile": 1234567990,
            "email": "abs@gmail.com",
            "gender": "female",
            "rating": 5
        }
    ],
    "pageable": {
        "sort": {
            "empty": false,
            "sorted": true,
            "unsorted": false
        },
        "offset": 0,
        "pageSize": 2,
        "pageNumber": 0,
        "unpaged": false,
        "paged": true
    },
    "last": false,
    "totalElements": 3,
    "totalPages": 2,
    "size": 2,
    "number": 0,
    "sort": {
        "empty": false,
        "sorted": true,
        "unsorted": false
    },
    "first": true,
    "numberOfElements": 2,
    "empty": false
}
```

Case 2 : In Appointment

AppointmentService :

```
public Page<Appointment> getAllAppointment(Integer pageNo, Integer pageSize, String SortBy) {
    Pageable pageable = PageRequest.of(pageNo, pageSize, Sort.by( ...properties: "appointmentId"));
    Page<Appointment> pagedResult = appointmentRepository.findAll(pageable);
    return pagedResult;
}
```

AppoinmentRepository :

```
Page<Appointment> findAll(Pageable pageable);
```

AppointmentController :

```
@GetMapping("/appointment")
    public ResponseEntity<Page<Appointment>> getAllAppointment(
            @RequestParam(defaultValue = "0") Integer pageNo,
            @RequestParam(defaultValue = "5") Integer pageSize,
            @RequestParam(defaultValue = "appointmentId") String sortBy) {
        Page<Appointment> appointments = appointmentService.getAllAppointment(pageNo, pageSize, sortBy);
        return ResponseEntity.ok()
                .header( headerName: "X-Total-Count", String.valueOf(appointments.getTotalElements()))
                .header( headerName: "X-Page-No", String.valueOf(appointments.getNumber()))
                .header( headerName: "X-Page-Size", String.valueOf(appointments.getSize()))
                .body(appointments);
}
```

Response :

(http://localhost:8090/api/v1/appointment?pageNo=0&pageSize=2&sortBy=id)


```
{
    "content": [
        {
            "createdDate": null,
            "createdBy": null,
            "lastModifiedDate": "2023-04-20T06:17:53.000+00:00",
            "lastModifiedBy": null,
            "appointmentId": 2,
            "user": {
                "createdDate": "2023-04-17T16:51:54.000+00:00",
                "createdBy": null,
                "lastModifiedDate": "2023-04-17T16:51:54.000+00:00",
                "lastModifiedBy": null,
                "id": 3,
                "username": "Srijansa",
```

```
                "password": "{bcrypt}$2a$10
$EOyBJvYxfApXdr5agTGwrerlcsKJMcFMeAzrbsJlHD
Ox/4EL44EtC",
                "category": "USER",
                "lastLoginDate": null,
                "roles": [],
                "enabled": true,
                "authorities": [],
                "accountNonLocked": true,
                "credentialsNonExpired": tr
ue,
                "accountNonExpired": true
            },
            "doctor": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "id": 3,
                "doctorSpecialization": {
                    "createdDate": null,
                    "createdBy": null,
                    "lastModifiedDate": nul
l,
                    "lastModifiedBy": null,
                    "id": 3,
                    "specializationName": "
Physician",
                    "specializationDescript
ion": "gthyj"
                },
                "hospital": {
                    "createdDate": null,
                    "createdBy": null,
                    "lastModifiedDate": nul
l,
                    "lastModifiedBy": null,
                    "hospitalId": 3,
                    "name": "sita",
                    "address": "Delhi"
                },
                "fullName": "Nasreen",
                "registrationNumber": "A1",
                "mobile": 1234567990,
                "email": "a@gmail.com",
                "gender": "female",
                "rating": 4
            },
            "payment": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "paymentId": 2,
                "amount": 12000.0,
                "status": "done",
                "mode": "online",
                "referenceId": "2"
            },
            "status": "done",
            "doctorSchedule": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "doctorScheduleId": 3,
                "doctor": {
                    "createdDate": null,
                    "createdBy": null,
                    "lastModifiedDate": nul
l,
                    "lastModifiedBy": null,
                    "id": 3,
                    "doctorSpecialization":
 {
                        "createdDate": null
,
                        "createdBy": null,
                        "lastModifiedDate":
 null,
                        "lastModifiedBy": n
ull,
                        "id": 3,
                        "specializationName
": "Physician",
                        "specializationDesc
ription": "gthyj"
                    },
                    "hospital": {
                        "createdDate": null
,
                        "createdBy": null,
                        "lastModifiedDate":
 null,
                        "lastModifiedBy": n
ull,
                        "hospitalId": 3,
                        "name": "sita",
                        "address": "Delhi"
                    },
                    "fullName": "Nasreen",
```

```
                    "registrationNumber": "
A1",
                    "mobile": 1234567990,
                    "email": "a@gmail.com",
                    "gender": "female",
                    "rating": 4
                },
                "startTime": "2023-04-
11T18:30:00.000+00:00",
                "endTime": "2023-04-
22T18:30:00.000+00:00",
                "capacity": 5
            }
        },
        {
            "createdDate": "2023-04-
20T08:40:13.000+00:00",
            "createdBy": null,
            "lastModifiedDate": "2023-04-
20T08:40:13.000+00:00",
            "lastModifiedBy": null,
            "appointmentId": 5,
            "user": {
                "createdDate": "2023-04-
17T16:48:24.000+00:00",
                "createdBy": null,
                "lastModifiedDate": "2023-
04-17T16:48:24.000+00:00",
                "lastModifiedBy": null,
                "id": 2,
                "username": "Srijansah",
                "password": "{bcrypt}$2a$10
$1nkmBQZgzQFKhC4x2qbBBe8C8E9Z408BA/elpc6Rcq
jE1aLgzoAWa",
                "category": "USER",
                "lastLoginDate": null,
                "roles": [],
                "enabled": true,
                "authorities": [],
                "accountNonLocked": true,
                "credentialsNonExpired": tr
ue,
                "accountNonExpired": true
            },
            "doctor": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "id": 3,
                "doctorSpecialization": {
                    "createdDate": null,
                    "createdBy": null,
                    "lastModifiedDate": nul
l,
                    "lastModifiedBy": null,
                    "id": 3,
                    "specializationName": "
Physician",
                    "specializationDescript
ion": "gthyj"
                },
                "hospital": {
                    "createdDate": null,
                    "createdBy": null,
                    "lastModifiedDate": nul
l,
                    "lastModifiedBy": null,
                    "hospitalId": 3,
                    "name": "sita",
                    "address": "Delhi"
                },
                "fullName": "Nasreen",
                "registrationNumber": "A1",
                "mobile": 1234567990,
                "email": "a@gmail.com",
                "gender": "female",
                "rating": 4
            },
            "payment": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "paymentId": 1,
                "amount": 10000.0,
                "status": "done",
                "mode": "online",
                "referenceId": "1"
            },
            "status": "done",
            "doctorSchedule": {
                "createdDate": null,
                "createdBy": null,
                "lastModifiedDate": null,
                "lastModifiedBy": null,
                "doctorScheduleId": 3,
                "doctor": {
                    "createdDate": null,
                    "createdBy": null,
                    "lastModifiedDate": nul
l,
```

```
                    "lastModifiedBy": null,
                    "id": 3,
                    "doctorSpecialization":
 {
                        "createdDate": null
,
                        "createdBy": null,
                        "lastModifiedDate":
 null,
                        "lastModifiedBy": n
ull,
                        "id": 3,
                        "specializationName
": "Physician",
                        "specializationDesc
ription": "gthyj"
                    },
                    "hospital": {
                        "createdDate": null
,
                        "createdBy": null,
                        "lastModifiedDate":
 null,
                        "lastModifiedBy": n
ull,
                        "hospitalId": 3,
                        "name": "sita",
                        "address": "Delhi"
                    },
                    "fullName": "Nasreen",
                    "registrationNumber": "
A1",
                    "mobile": 1234567990,
                    "email": "a@gmail.com",
                    "gender": "female",
                    "rating": 4
                },
                "startTime": "2023-04-
11T18:30:00.000+00:00",
                "endTime": "2023-04-
22T18:30:00.000+00:00",
                "capacity": 5
            }
        ],
        "pageable": {
            "sort": {
                "empty": false,
                "sorted": true,
                "unsorted": false
            },
            "offset": 0,
            "pageSize": 2,
            "pageNumber": 0,
            "unpaged": false,
            "paged": true
        },
        "last": false,
        "totalElements": 4,
        "totalPages": 2,
        "size": 2,
        "number": 0,
        "sort": {
            "empty": false,
            "sorted": true,
            "unsorted": false
        },
        "first": true,
        "numberOfElements": 2,
        "empty": false
}
```


Case 3 : In DoctorSpecialization

DoctorSpecializationService :

```
public Page<DoctorSpecialization> getDoctorSpecialization(Integer pageNo, Integer pageSize, String SortBy) throws ZumplyException{
    Pageable pageable = PageRequest.of(pageNo, pageSize, Sort.by( ...properties: "id"));
    Page<DoctorSpecialization> doctorSpecializationList = DoctorSpecializationRepository.findAll(pageable);
    if (doctorSpecializationList.isEmpty()){
        throw new ZumplyException(ZumplyErrorCode.TABLE_IS_EMPTY.getErrorCode(), ZumplyErrorCode.TABLE_IS_EMPTY.getErrorMessage());
    }
    return doctorSpecializationList;

}
```

DoctorSpecializationRepository :

```
Page<DoctorSpecialization> findAll(Pageable pageable);
```

DoctorSpecializationController :

```
@GetMapping("/category")
public ResponseEntity<?> getDoctorSpecialization(@RequestParam(defaultValue = "0") Integer pageNo,
                                                 @RequestParam(defaultValue = "5") Integer pageSize,
                                                 @RequestParam(defaultValue = "id") String sortBy,
                                                 @Valid DoctorSpecialization DoctorSpecialization) throws ZumplyException{
    try{
        Page<DoctorSpecialization> doctorSpecializationList = DoctorSpecializationService.getDoctorSpecialization(pageNo, pageSize, sortBy);
        return ResponseEntity.ok()
                .header( headerName: "X-Total-Count", String.valueOf(doctorSpecializationList.getTotalElements()))
                .header( headerName: "X-Page-No", String.valueOf(doctorSpecializationList.getNumber()))
                .header( headerName: "X-Page-Size", String.valueOf(doctorSpecializationList.getSize()))
                .body(doctorSpecializationList);
    }
    catch (ZumplyException e) {
        ErrorResponse errorResponse = new ErrorResponse(e.getErrorCode(), e.getErrorMessage());
        return ResponseEntity.ok().body(errorResponse);
    }
}
```

Response :

(http://localhost:8090/api/v1/category?pageNo=0&pageSize=2&sortBy=id)


```
{
    "content": [
        {
            "createdDate": null,
            "createdBy": null,
            "lastModifiedDate": null,
            "lastModifiedBy": null,
            "id": 1,
            "specializationName": "Den
tist",
            "specializationDescription
": "asdsdsfg"
        },
        {
            "createdDate": null,
            "createdBy": null,
            "lastModifiedDate": "2023-
04-21T09:41:00.000+00:00",
            "lastModifiedBy": null,
            "id": 2,
            "specializationName": "Den
tist  RM",
            "specializationDescription
": "MBBS specialist"
        }
    ],
    "pageable": {
        "sort": {
            "empty": false,
            "sorted": true,
            "unsorted": false
        },
        "offset": 0,
        "pageSize": 2,
        "pageNumber": 0,
        "unpaged": false,
        "paged": true
    },
    "last": false,
    "totalElements": 14,
    "totalPages": 7,
    "size": 2,
    "number": 0,
    "sort": {
        "empty": false,
        "sorted": true,
        "unsorted": false
    },
    "first": true,
    "numberOfElements": 2,
    "empty": false
}
```